

Table 1 · Minimal instance repair

Method	R APall	R AP50	R mIoU	CF APall	CF AP50	CF mIoU	CF ΔAP G1	CF ΔAP D2
IR00 D2	11.74	24.50	29.69	8.31	17.96	19.00	0.08	0.00
IR01 G1	12.39	26.06	30.27	8.23	17.83	19.02	0.00	-0.08
IR02 Nearest attach	12.20	25.45	30.23	8.23	17.84	19.02	0.00	-0.07
IR03 Evidence attach	12.39	26.06	30.27	8.23	17.83	19.02	0.00	-0.08
IR04 Direct group	12.39	26.06	30.27	8.23	17.83	19.02	0.00	-0.08
IR05 Verified repair	12.39	26.06	30.27	8.23	17.83	19.02	0.00	-0.08
IR06 AnyUp reread	9.94	21.92	27.44	8.62	19.02	20.92	0.39	0.31
IR07 Boundary stable	10.63	23.29	28.28	8.53	18.72	20.48	0.30	0.22
IR08 Combination	10.63	23.29	28.28	8.53	18.72	20.48	0.30	0.22

Official dataset pools (%), fixed surface, one exclusive partition. Deltas are percentage points; all five metrics in source data.

Table 2 · Minimal instance repair

Pair	Cohort	ΔAP_{all}	ΔAP_{50}	$\Delta mIoU$	Ops A/B	Geom50 +/-	Geom75 +/-	GT50 +/-	GT75 +/-	W→R	R→W
IR03–IR02	Replica	0.19	0.61	0.05	0/25	1/0	1/0	1/0	1/0	0	0
IR05–IR04	Replica	0.00	0.00	0.00	1/1	0/0	0/0	0/0	0/0	0	0
IR07–IR06	Replica	0.69	1.37	0.84	0/0	0/0	0/0	1/0	0/0	1	0
IR08–IR05	Replica	-1.75	-2.77	-1.99	1/1	0/0	0/0	0/3	0/1	0	3
IR08–IR07	Replica	0.00	0.00	0.00	1/0	0/0	0/0	0/0	0/0	0	0
IR03–IR02	CF18	-0.00	-0.01	-0.00	6/93	1/0	0/0	0/0	0/0	0	0
IR05–IR04	CF18	0.00	0.00	-0.00	1/6	0/0	0/0	0/0	0/0	0	0
IR07–IR06	CF18	-0.09	-0.30	-0.44	0/0	0/0	0/0	0/1	0/0	0	1
IR08–IR05	CF18	0.30	0.88	1.45	1/1	0/0	0/0	5/0	2/0	5	0
IR08–IR07	CF18	0.00	0.00	0.00	1/0	0/0	0/0	0/0	0/0	0	0

Prespecified pairs. Geom: class-agnostic matching; GT: released class-aware matching. W/R: geometrically matchable original P supports. Full thresholds and label changes in supplement.

Table 3 · Minimal instance repair

Method	R APall	R mIoU	CF APall	CF mIoU	s/scene	Alloc GiB	Reserv GiB	FC/call	QK/call	Regions/call	Obs/call
IR00 D2	11.74	29.69	8.31	19.00	—	—	—	—	—	—	—
IR01 G1	12.39	30.27	8.23	19.02	16.23	1.89	2.21	4.50	0.00	0.00	0.00
IR02 Nearest attach	12.20	30.23	8.23	19.02	—	—	—	—	—	—	—
IR03 Evidence attach	12.39	30.27	8.23	19.02	—	—	—	—	—	—	—
IR04 Direct group	12.39	30.27	8.23	19.02	—	—	—	—	—	—	—
IR05 Verified repair	12.39	30.27	8.23	19.02	85.57	1.89	2.21	4.50	0.00	0.00	32.00
IR06 AnyUp reread	9.94	27.44	8.62	20.92	—	—	—	—	—	—	—
IR07 Boundary stable	10.63	28.28	8.53	20.48	81.10	4.40	6.65	20.12	16.75	80.50	32.00
IR08 Combination	10.63	28.28	8.53	20.48	106.15	4.40	6.65	20.12	16.75	80.50	32.00

Fresh paired cold calls, 8 Replica scenes × 2 rounds per measured arm. Required models resident; peaks are maxima, time is the mean. Unmeasured: —.